

When Gestures and Words Synchronize: Exploring A Human Lecturer's Multimodal Interaction toward the Design of Embodied Pedagogical Agents

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Figure 1: Three Examples of Spoken Words Aligning with Contextual Gestures. Left: The lecturer's left-hand moves from left to right to show two separate spaces; Middle: The lecturer's left-hand moves toward his chest; Right: His left hand draws a continuity circle.

ABSTRACT

Embodied Pedagogical Agents (EPAs) possess significant potential for using multimodal interactions to effectively convey social and educational information. While previous studies have examined the impact of EPAs on learning performance, the incorporation of meaningful gestures into EPAs remains relatively unexplored. This study employs a case study approach and systemic functional-multimodal discourse analysis to investigate the multimodal interactions of a human lecturer during teaching and explores the relationship between in-class meaningful gestures and corresponding words. We analyze the alignment between gestures, words, and contextual meanings, with a specific focus on personal pronouns, adjectives, adverbs, and verbs. The findings contribute to the understanding of lecturers' multimodal orchestration in classroom settings. Additionally, we provide a preliminary design guideline for EPAs that integrates meaningful word-synchronized gestures, using multimodal information to enhance students' embodied learning experiences. This

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research contributes to the advancement of embodied pedagogical agent design and the broader field of multimodal interaction.

CCS CONCEPTS

• **Applied computing** → Education; E-learning.

KEYWORDS

Pedagogical agent, Gesture, Word synchronized, Multimodal interaction, Human behavior

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1 INTRODUCTION

Embodied Pedagogical Agents (EPAs) have immense potential for using multimodal interactions to convey both verbal and visual social and educational information [23]. Numerous studies have evaluated the impact of EPAs on students' learning performance [4, 7, 13, 20, 22]. Recent research has shown that PAs with high levels of human-like behaviors positively influence students' perception of social presence [31, 33]. Additionally, studies have indicated that PAs incorporating meaningful gestures, such as deictics, iconics, and metaphors, can enhance students' learning outcomes

[8, 30, 35]. Scholars have noted that teachers across different regions use gestures during instructional analogies to foster students' abstract thinking and enhance comprehension [26, 27]. Therefore, it is crucial to investigate the multimodal interactions of human lecturers during in-class teaching to inform the design of EPAs that incorporate meaningful gestures.

While some researchers have explored co-speech gesture generation for designing gestural agents, with focuses on speech rhythm [1, 16, 19] and word embedding [34], the incorporation of meaningful word-synchronized gestures into EPAs remains largely unexplored. Examining the natural movements of human lecturers and their alignment with speech in real-life teaching scenarios holds promise for designing EPAs that incorporate meaningful word-synchronized gestures, ultimately leading to improved student learning effectiveness.

This study presents an initial understanding of the alignments between lecturers' gestures and corresponding words and their parts of speech, specifically focusing on personal pronouns, adjectives, adverbs, and verbs. The findings contribute in the following ways: 1) It provides empirical insights into the alignment between gestures, words, and their contextual meanings. 2) It offers a preliminary exploration of word-synchronized gestures used by human lecturers. 3) It presents the design of word-synchronized gestural EPAs that use integrated multimodal information to convey contextual meanings and enhance embodied learning experiences for students.

2 RELATED WORK

2.1 Gesture

Gesture, as an essential form of multimodal interaction, reflects human thought and enhances language semantics [10, 15, 23]. Previous studies on human gestures have primarily focused on three key attributes: gesture types, continuum, and meanings. McNeill identified five types of gestures: beats, cohesives, deictics, metaphors, and iconics [21]. Furthermore, gesture exhibits a continuum process, involving four stages: pre-preparation position, pre-stroke hold, stroke, and post-stroke hold, which convey a chronological system of gestures' trajectory [15]. The study of gesture types and continuum has been adopted in the field of human-computer interaction to understand the representations of human cognitive processes and behavioral patterns [24, 25, 29]. Lim formulated a social-related classification of gestures [24], which includes communicative gestures reflecting self-representation and language correlation, as well as performative gestures [18].

This study draws on the classifications of gestures by McNeill [21], Kendon [34], and Lim [18] to identify word-synchronized gestures of a lecturer in a real-life teaching environment, aiming to gain an initial understanding of the in-class multimodal interaction exhibited by human lecturers.

2.2 Word

Verbal language serves as the primary communication medium between lecturers and students [28]. It has been observed that human communicative signals are often accompanied by the integration of spoken language and gestures [2]. Neuroscientific research has indicated that human comprehension and pronunciation of words,

particularly their meanings and part-of-speech (PoS, e.g., adjectives, adverbs, verbs, etc.), rely on sensorimotor systems rather than purely linguistic systems, which primarily function in abstract word awareness [5, 9, 14]. The meanings of words that relate to images and pictorial scenarios often elicit iconic gestures. For instance, these gestures, such as mimicking drinking when engaging in a conversation about thirst, serve as vivid visual representations of the concepts they embody [32]. The alignment of words with gestures holds great potential for uncovering the multimodal interaction exhibited by human lecturers. However, research focusing on analyzing the PoS of words in lectures and their corresponding gestures is relatively scarce. Exploring word usage within linguistic contexts shows promise in understanding the gestures employed by lecturers.

3 A CASE STUDY

The case study approach shows potential in revealing multimodal interactions, specifically the alignment between a lecturer's gestures and words, and it offers an initial design guideline for EPAs incorporating word-synchronized gestures. To analyze the meanings conveyed through linguistic and gestural elements, this study adopts a lens of Systemic Functional Theory [17]. By employing systemic functional-multimodal discourse analysis, the study aligns gesture and action type analysis with content analysis, thus exploring the relationship between gestures and words.

3.1 Subject, Research Site, and Procedure

Based on the characteristics of natural behaviors exhibited by instructors in a real-life teaching setting, we conducted an initial study in a typical university classroom (a square-shaped room with the lecturer standing at the front). To ensure high-quality in-class gesture data, specific requirements were established for participant recruitment. The recruited instructor had to meet two criteria: 1) actively using gestures to capture students' attention and enhance comprehension, and 2) having noticeable and substantial in-class gestures as reported by students.

A male participant, identified as George (a pseudonym), was recruited for the study. George is a university lecturer with ten years of teaching experience at the undergraduate level, specializing in the field of design. He met the recruitment requirements and willingly participated in the study. The observed course was George's class on user experience design.

During the class session, a camera was positioned at a 3/4 angle approximately 3 meters away from the instructor. One of our researchers was present to observe the instructor's actions and ensure uninterrupted operation of the camera, without disrupting the regular teaching process. Around thirty university students were present in the classroom, attentively listening to George's lecture. The observation and video recording lasted for one hour, encompassing George's speech, non-verbal behaviors, and the entirety of the teaching process.

3.2 Materials

The study used the multimodal discourse analysis software ELAN. Four tiers were created, recorded, annotated, and analyzed, encompassing lecturer action types, gesture types and continuum, as well

Table 1: A Description of Gesture Types and Continuum, and Action Types from George's Lecture Video Recording

Gesture Type		Beats	Cohesives	Deictics	Iconics	Metaphorics	Null	Total
	Fre.	1040	560	461	83	284	374	2802
	%	37.1	20.0	16.5	3.0	10.1	13.3	100.0
Duration (ms)	Ave.	1693	863.53	965.62	1575.3	1356.25		
	Max.	23970	9100	15060	7800	15770		
	Total	1760715	483577	445150	130750	375680		3195872
Action type		Indexical	Presenting	Representing	Null			Total
		Action	Action	Action				
	Fre.	1640	202	731	229			2802
	%	58.5	7.2	26.1	8.2			100.0
Representing and Presenting Action		Behavior	Interpersonal	Material	Null			Total
		Process	Meanings	Process				
	Fre.	108	1987	511	196			2802
	%	3.9	70.9	18.2	6.9			100.0
		Pre-preparation position	Prestroke hold	Stroke	Post-stroke hold	Null		Total
Gesture	Fre.	261	239	1655	347	300		2802
Continuum	%	9.3	8.5	61.0	12.4	10.7		100.0

Null= George doesn't show any bodily language; Fre. = the frequency of certain gesture type in George's lecture; %=percentage of certain gesture type in George's lecture; Ave. = the average duration of certain gesture type in George's lecture; Max. = the maximum duration of certain gesture type in George's lecture.

as speech content, to examine lecturers' social involvement. An online transcription service was employed to transcribe the spoken words in George's lecture video, resulting in a total of 13,659 words that were manually reviewed for accuracy. During the manual review, the use of personal pronouns, adjectives, adverbs, and verbs was specifically annotated. Data analysis involved the use of statistical software to calculate word frequencies and their co-occurrence with gestures.

3.3 Lecturer's Gestures and Words Synchronize

3.3.1 George's In-class Gesture Description. We have annotated the frequency and duration (i.e., total, average, maximum) of the five gesture types in George's lecture video recording, as well as the frequency and percentage of gesture continuum and action types, using SPSS. The data is presented in Table 1. George performed the beats gesture 1040 times, with a maximum duration of 23970ms and an average duration of 1693ms. He used cohesive gestures 560 times, with a maximum duration of 9100ms and an average duration of 863.53ms. Deictic gestures occurred 461 times, with a maximum duration of 15060ms and an average duration of 965.62ms. Iconic gestures were used 83 times, with a maximum duration of 7800ms and an average duration of 1575.3ms. Metaphoric gestures were observed 277 times, with a maximum duration of 15770ms and an average duration of 1356.25ms. Additionally, George used non-bodily movements 374 times.

The data reveals that George extensively used beats gestures (37.1%) in terms of both frequency and duration. Cohesive gestures occurred frequently and had shorter durations, suggesting their common usage as auxiliary gestures. Deictic gestures had a relatively high usage rate (16.5%), and metaphoric gestures (10.1%) also

showed significant frequency. George used deictic and metaphoric gestures relatively more. Although he used iconic gestures less frequently, their average duration was longer. Contiguous stroking gestures (61%) were highly observed, while gestures in the gesture preparation stage (pre-preparation position: 9.3%; pre-stroke hold: 8.5%) were less frequent. Indexical action (58.5%) and representing action (26.1%) were highly exhibited, with interpersonal meanings (70.9%) showing a significant presence.

During George's one-hour lecture, he intentionally interacted with students and used abundant nonverbal language to complement his verbal explanations. He engaged in numerous social actions while teaching. George effectively used gestures primarily to interact with students and the physical teaching environment. Based on the results of gesture frequency and duration in the classroom, we consider the analysis of George's gestures feasible for gaining an initial understanding of intentionally embodied interaction by human lecturers.

3.3.2 The Alignments between Gestures and Words. Based on George's speech, we found he used the pronoun "I" 41 times, "we" 30 times, and "you" 76 times, indicating a relatively equal distribution of first-person and second-person pronouns in his language use during the teaching process. He frequently employed the adjectives "different" (39 times, 0.29%), "same" (5 times, 0.04%), and "more" (14 times, 0.1%), as well as adjectives like "new" (11 times, 0.08%), "first" (8 times, 0.06%), and "small" (7 times, 0.05%). These adjectives were primarily associated with nouns related to abstract concepts such as titles, projects, time, and groups. They consistently appeared alongside George's beat, metaphoric, and iconic gestures. The purpose

of using these adjectives was to externalize the lecturer's perception by employing visually comprehensible gestures to enhance the understanding of abstract content.

However, this alignment between adjectives and gestures was not observed in George's use of adverbs. Lexis like "so" (41 times, 0.3%), "not" (36 times, 0.26%), and "then" (15 times, 0.11%) frequently occurred, presumably contributing to the cohesion of sentence structure. Other adverbs like "really" (9 times, 0.07%), "better" (2 times, 0.01%), and "much" (6 times, 0.04%) were used to express emphasis and aligned with beat and cohesive gestures.

The use of verbs in the teaching process was actively relevant with meaningful gestures. In addition to common verbs like "be/is/was/are/am/been" (167 times, 1.22%), "do/doing/does" (36 times, 0.26%), and "have" (30 times, 0.22%), verbs like "see" (21 times, 0.15%), "come" (8 times, 0.06%), "put" (8 times, 0.06%), and "follow" (9 times, 0.07%) were identified. The use of these verbs is able to visualize bodily movements and scenarios and frequently co-occurred with metaphoric and iconic gestures, revealing significant and relevant patterns.

3.4 Four Patterns of the Lecturer's Part-of-speech- Lexis -Gesture Alignments

Based on the typical part-of-speech-word-gesture alignments observed in George's one-hour lecture (refer to Table 2), four main patterns of word-synchronized gestures were identified.

The alignment between personal pronouns and deictic and beat gestures. The data indicates that George primarily used beat gestures when speaking from a first-person perspective (e.g., "I can create..."). Conversely, he employed numerous deictic gestures when addressing second- and third-person perspectives (e.g., "but you have time...").

The alignment between adjectives and metaphoric, iconic, and beat gestures. When the meaning of the adjectives pertained to space and size, which are spatial and constructive approaches (e.g., *different*, *more*, *small*, *following*, *whole*, *little*), George used metaphoric and iconic gestures. Generally, the use of adjectives prompted George to employ beat gestures.

The alignment between adverbs and cohesive, beat, and iconic gestures. George seemingly used adverbs to emphasize the subject words and logical aspects of the speech context. When emphasizing, he used beat gestures (e.g., "even have all these patients..."). Furthermore, when explaining the logic, he predominantly employed cohesive gestures (e.g., "...so it's not just two factors..."). Additionally, he employed adverbs with spatial and constructive meanings (e.g., *together*, *back*) in conjunction with metaphoric and iconic gestures.

The alignment between verbs and metaphoric and beat gestures. A strong relationship was observed between speaking verbs associated with pictorial actions (e.g., *see*, *follow*, *come*, *put*, *read*, *analyze*, etc.) and contextual gestures, including metaphoric and iconic gestures. For instance, George's left-hand moves from left to right to show two separate spaces, when he said "...have *different* titles and *different* ways of doing" (shown in Figure 1 left). When he mentioned "when you put that *together* ...", his left-hand moves toward his chest (Figure 1 middle). His left hand

draws a continuity circle, while he said "I'm *processing*. I know what you can. ...". (Figure 1 right).

4 TOWARDS A DESIGN OF WORD-SYNCHRONIZED GESTURAL PEDAGOGICAL AGENTS

The study aims to investigate the multimodal interaction of human lecturers in real-life classrooms to inform the development of Embodied Pedagogical Agents (EPAs) with word-synchronized gestures. The findings reveal a strong correlation between lecturers' spoken words and specific types of gestures. When employing personal pronouns, lecturers establish social connections through second- and third-person perspectives, often accompanied by spatial-oriented gestures like deictic gestures. Conversely, when providing self-related information, lecturers tend to focus on self-comprehension, leading to the use of beat gestures. These findings partially support McNeill's concept of dual viewpoints in gestures [21].

In the use of adjectives, adverbs, and verbs, lecturers intentionally emphasize attributes to highlight speech content. Rhetoric-oriented adjectives, adverbs, and beat gestures are performed accordingly. When employing words for sentence coherence, such as "so", "then", "following", and "going to", lecturers use beat or cohesive gestures. This finding aligns with McNeill's gesture study [21]. Additionally, contextual gestures, including metaphoric and iconic gestures, are extensively employed by lecturers when discussing speech content with spatial, constructive, and pictorial-action meanings. This result is consistent with neuroscientific studies discussed in previous sections [5, 14, 32]. EPA designers should consider the four patterns of the lecturer's part-of-speech-lexis-gesture alignments as a fundamental design guideline for the development of word-synchronized gestural pedagogical agents.

5 DISCUSSION AND CONCLUSION

Our results offer preliminary insights into the alignment between a lecturer's meaningful gestures and words in a classroom setting. Gesture has evidenced that it obtains great potential in enhancing lecturers' self-comprehension during the teaching process and improving students' learning outcomes [3, 6, 11, 12]. Several studies have explored gesture generation algorithms that demonstrate excellent accuracy and usability [1, 16, 19, 34]. Our findings hold promise in bridging the previous research and providing a design guideline for creating EPAs that effectively integrate meaningful word-synchronized gestures.

Implication. This study provides a foundation for the design and development of the EPAs that effectively integrate meaningful word-synchronized gestures, aiming to enhance student learning effectiveness and enrich the educational experience.

- This empirical exploration provides insights into a lecturer's multimodal interaction, the integration of verbal and visual information, in educational settings.
- In addition to enhancing human-like behaviors, EPAs can be designed with meaningful word-synchronized gestures to enrich the multimodal information of conveying educational meaning. By

Table 2: Four Part-of-speech-Lexis-Gesture alignment Patterns Based on George's Gestures and Speech Content

Word			Gesture	
Part-of-speech	Lexis	Lexis in George's Speech Context	Gesture Type	Gesture Description
Personal Pronoun	I	<i>it's actually when I . .</i>	Beats	George's left-hand suddenly moves upward.
	my	<i>that's something different from my</i>	Beats	He suddenly spreads his arms.
	we	<i>there we go. Other things to respond to steel is a much</i>	Beats	His left-hand suddenly moves upward.
Adjectives	you	<i>when you think about the topic, was a good</i>	Deictics	His left-hand points toward students.
	your	<i>is your picture understanding our teams</i>	Deictics	His left-hand points toward students.
	they	<i>this is what they were trying to do</i>	Deictics	His left-hand points to the left space.
	different	<i>have different titles and different ways of doing</i>	Metaphorics	His left-hand moves from left to right to show two separate spaces.
	same	<i>the same activity is possible</i>	Beats	He suddenly spreads his arms.
Adverbs	more	<i>spend more time to give you</i>	Metaphorics	He raises his left-hand and compares to the lower space.
	small	<i>confusion small observation to consider</i>	Metaphorics	He brings his hands together.
	whole	<i>to see the whole concept</i>	Metaphorics	He draws a large circle with both arms.
	little	<i>switch a little bit instead of improving</i>	Iconics	He presses his fingers together.
	so	<i>So what's the process of analyzing</i>	Cohesives	He moves hands slightly from right to left.
	also	<i>overview of the subject also the structure</i>	Cohesives	He moves hands slightly from right to left.
	even	<i>even have all these patients present in a way</i>	Beats	He suddenly spreads his arms.
	together	<i>when you put that together</i>	Iconics	His left-hand moves toward his chest.
	better	<i>So: the better it is better, but you have time</i>	Beats	His hands suddenly move up and down.
	back	<i>you see, going back to follow expect any</i>	Iconics	His left-hand moves from the front to slightly back.
Verbs	is	<i>to calculate is going to be</i>	Beats	His hands suddenly move up and down.
	doing	<i>on the ways of doing it</i>	Beats	His hands suddenly move up and down.
	see	<i>most of the time, for us to see the whole concept</i>	Metaphorics	He draws a large circle with both arms.
	analyzing	<i>analyzing boxes, it's like a process of summarizing.</i>	Metaphorics	With his left hand, he makes the form of holding a pen and moves through the air.
	enhanced	<i>enhanced context, it's also different people</i>	Metaphorics	He raises his left-hand and compares to the lower space.
	processing	<i>I'm processing. I know what you can</i>	Metaphorics	His left hand draws a continuity circle.
	provide	<i>provide them with information on the ways of doing it</i>	Metaphorics	He spreads his left palm to the students.

incorporating meaningful gestures, EPAs can provide more precise and accurate representations of the intended message, thereby enhancing the effectiveness of communication with students.

c) This study explores word-gesture synchronization not only of word meanings but also of word parts-of-speech, providing gestures that are more closely aligned with the temporal structure of language. This perspective on the structural aspect of language contributes to more accurate and synchronized gesture cues, creating a more cohesive and meaningful interaction between EPAs and students.

d) The study highlights the importance of considering gestures in conjunction with words. It recognizes that human communication involves the integration of spoken language and gestures, and that the alignment between words and gestures holds potential for understanding multimodal interactions in educational contexts. This

insight opens avenues for further research in analyzing the part-of-speech of words and their corresponding gestures, contributing to a deeper understanding of lecturers' gesture usage.

Future Work. This research is ongoing, and we will continue to analyze the multimodal interaction of human lecturers and evaluate the usability of the gesture design framework through laboratory experiments. In the future development phase, we will engage lecturers in the co-design of gestural EPAs. Ultimately, we will create EPAs with meaningful word-synchronized gestures and assess their impact on students' learning performance and perception. Additionally, it is important to consider factors such as students' perception of the lecturer's gestures and cultural differences in human gesture use, which may influence the embodied movements of lecturers in the classroom. Therefore, further research should be conducted to explore these factors.

Limitation. This study is based on a case study, and the speech and gesture behaviors of individuals may influence the results, potentially leading to sample bias. The analysis in this study primarily focuses on four parts of speech, but other parts of speech, such as nouns and adverbs, may also interact and have an impact. In future research, a mixed-method analysis will be employed, incorporating a broader range of vocabulary to offer a more comprehensive framework for gesture design.

Conclusion. Our study expands on EPA design that uses a word as a unit and explores its contextual meanings to provide students with a more authentic and embodied learning experience. It differs from previous EPA studies in that we found the alignments between words and gestures and uncovered word-synchronized gestures rather than speech-rhythm-based gestures. The analysis results show the importance of EPA studies in the HCI community.

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